

CLOUD COMPUTING-II

Unit-I

Presented By

Mr.U.Arun,
Assistant Professor,
Dept. of MCA,

Parul Institute of Engineering and Technology

Introduction to Networking in the Cloud

7 Layers of the OSI Model

Application

- End User layer
- HTTP, FTP, IRC, SSH, DNS

Presentation

- Syntax layer
- SSL, SSH, IMAP, FTP, MPEG, JPEG

Session

- Synch & send to port
- API's, Sockets, WinSock

Transport

- End-to-end connections
- TCP, UDP

Network

- Packets
- IP, ICMP, IPsec, IGMP

Data Link

- Frames
- Ethernet, PPP, Switch, Bridge

Physical

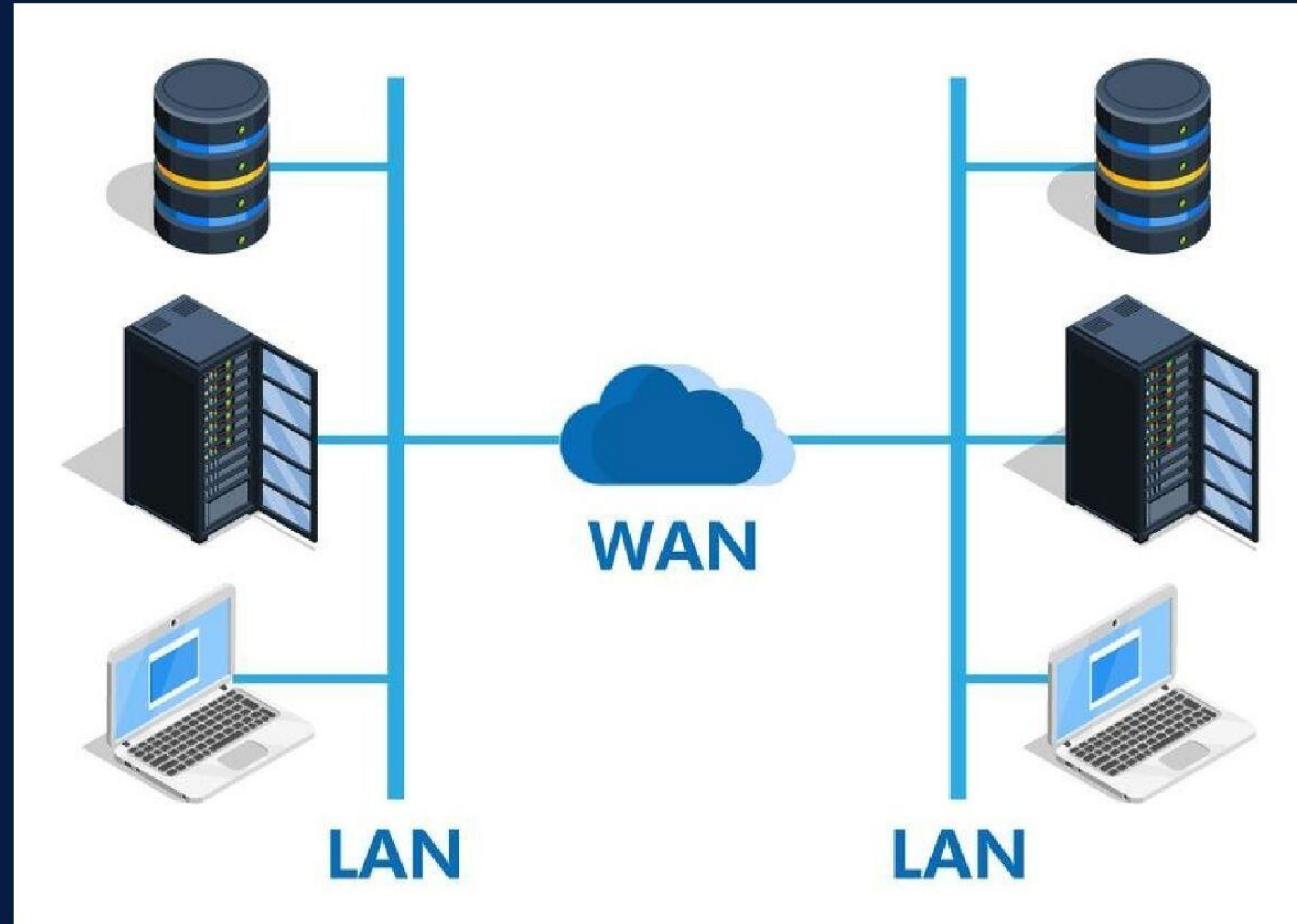
- Physical structure
- Coax, Fiber, Wireless, Hubs, Repeaters

LA

A **Local area network (LAN)** is a collection of devices connected together in one physical location, such as a building, office or home.



WAN

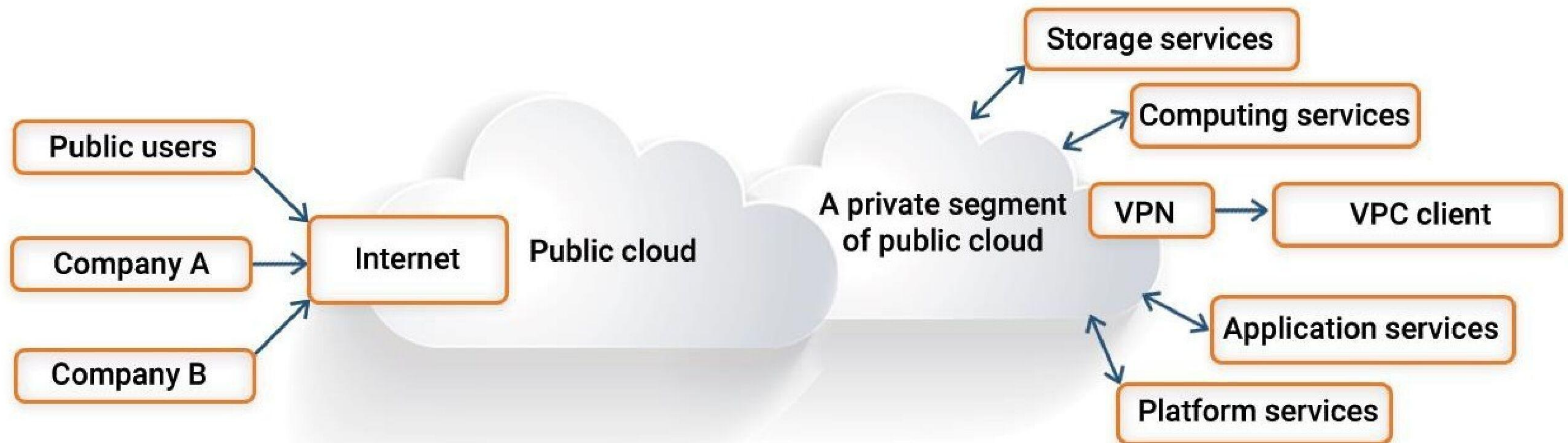


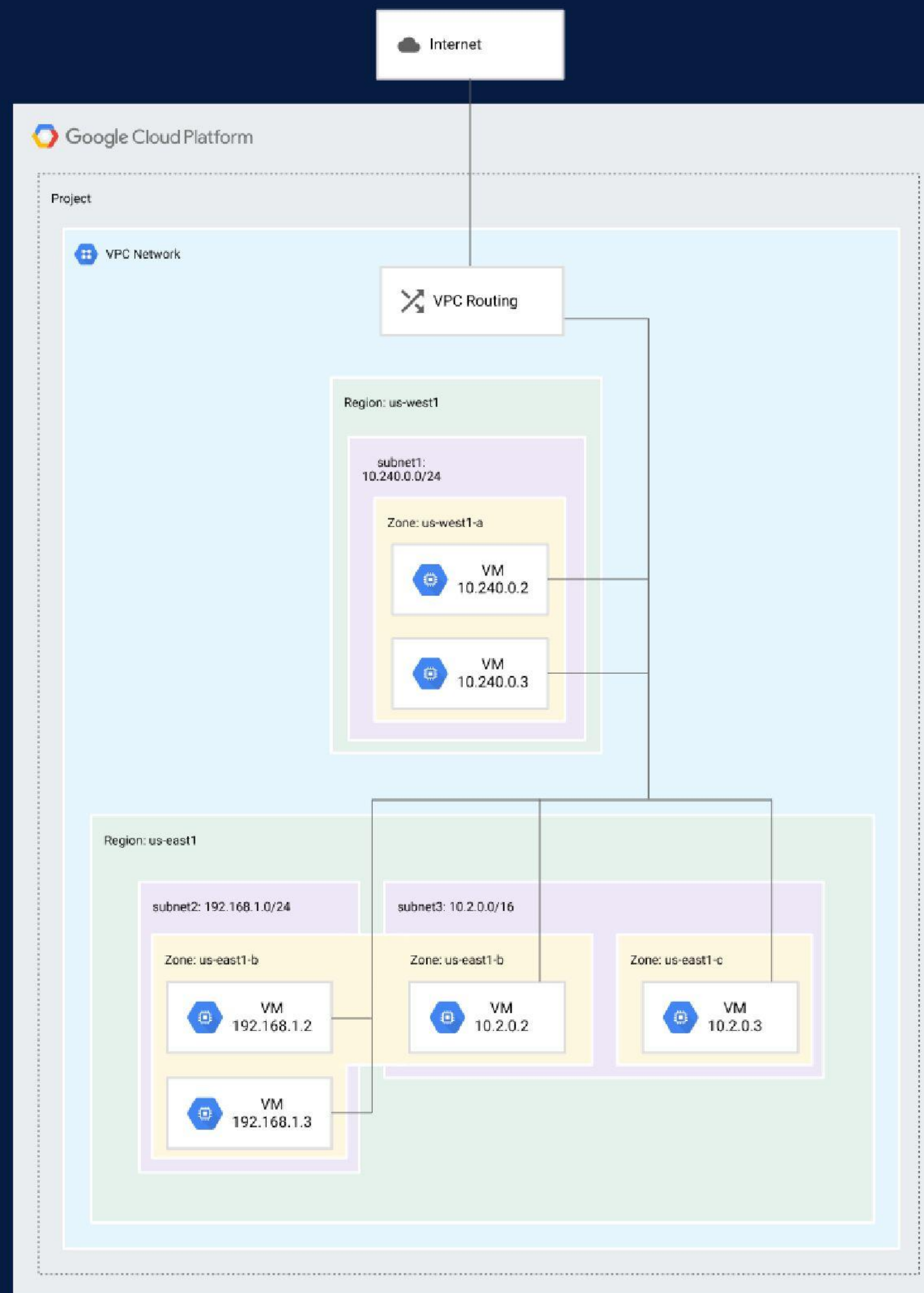
A wide-area network (WAN) is the technology that connects your offices, data centers, cloud applications, and cloud storage together.

VPC

Virtual private cloud (VPC) is a type of hybrid cloud deployment. VPC is a private cloud within a public cloud (remote provider), where users can perform regular operations as in a private cloud. These functionalities can include storing data, running programming code, or even hosting websites. In a way, a virtual cloud is a combination of a private and public cloud.

VIRTUAL PRIVATE CLOUD





- Subnet1 is defined as 10.240.0.0/24 in the us- west1 region.Two VM instances in the us-west1-a zone are in this subnet. Their IP addresses both come from the available range of addresses in subnet1.
- Subnet2 is defined as 192.168.1.0/24 in the us- east1 region.Two VM instances in the us-east1-b zone are in this subnet. Their IP addresses both come from the available range of addresses in subnet2.
- Subnet3 is defined as 10.2.0.0/16, also in the us- east1 region.One VM instance in the us-east1-b zone and a second instance in the us-east1-c zone are in subnet3, each receiving an IP address from its available range. Because subnets are regional resources, instances can have

IP addresses

Resources such as VM instances and load balancers have IP addresses in Google Cloud. These IP addresses let Google Cloud resources communicate with other resources in Google Cloud, in on-premises networks, or on the public internet.

External IP address

Internal IP address

Private IP address

Public IP address

External IP address

External IP addresses are publicly advertised, meaning they are reachable by any host on the internet. External IP addresses must be publicly routable IP addresses. Resources with external IP addresses can communicate with the public internet.

Internal IP address

Internal IP addresses cannot be reached from the internet, and are not publicly routable. Internal IPv4 addresses can be private IPv4 addresses or they can be privately used public IPv4 addresses. For a list of valid internal IPv4 addresses, see [Valid IPv4 ranges](#). Internal IPv6 addresses are unique within Google Cloud

Private IP address

Private IP addresses are addresses that cannot be routed on the internet. For a list of private IPv4 ranges, see the entries for Private IP address ranges in the valid internal IPv4 address ranges table.

Public IP address

Public IP addresses are internet routable. In Google Cloud, external IPv4 and IPv6 addresses are always public IP addresses. You can also use public IPv4 addresses as internal addresses when you configure the primary or secondary IPv4 address range of a subnet in your VPC network. These addresses are referred to as privately used public IP addresses.

Routes and Firewall Rules in the Cloud

Virtual Private Cloud (VPC) firewall rules apply to a given project and network. If you want to apply firewall rules to multiple VPC networks in an organization.

VPC firewall rules let you allow or deny connections to or from virtual machine (VM) instances in your VPC network. Enabled VPC firewall rules are always enforced, protecting your instances regardless of their configuration and operating system.

Firewall rules in Cloud

When you create a VPC firewall rule, you specify a VPC network and a set of components that define what the rule does. The components enable you to target certain types of traffic, based on the traffic's protocol, destination ports, sources, and destinations.

In addition to firewall rules that you create, Google Cloud has other rules that can affect incoming (ingress) or outgoing (egress) connections

VPC firewall rules have the following characteristics:

Each firewall rule applies to incoming (ingress) or outgoing (egress) connections, not both. For more information, see [direction of connection](#).

Firewall rules support IPv4 connections. IPv6 connections are also supported in VPC networks that have IPv6 enabled. When specifying a source or destination for an ingress or egress rule by address, you can specify IPv4 or IPv6 addresses or blocks in CIDR notation.

Each firewall rule can contain either IPv4 or IPv6 ranges, but not both.

Each firewall rule's action is either allow or deny. The rule applies to connections as long as it is enforced. For example, you can disable a rule for troubleshooting purposes.

When you create a firewall rule, you must select a VPC network. While the rule is enforced at the instance level, its configuration is associated with a VPC network. This means that you cannot share firewall rules among VPC networks, including networks connected by VPC Network Peering or by using Cloud VPN tunnels.

VPC firewall rules are stateful

When a connection is allowed through the firewall in either direction, return traffic matching this connection is also allowed. You cannot configure a firewall rule to deny associated response traffic.

Return traffic must match the 5-tuple (source IP, destination IP, source port, destination port, protocol) of the accepted request traffic, but with the source and destination addresses and ports reversed.

Google Cloud associates incoming packets with corresponding outbound packets by using a connection tracking table.

IPv4 connections support TCP, UDP, SCTP, and ICMP protocols. IPv6 connections support TCP, UDP, SCTP, and ICMPv6 protocols.

Google Cloud implements connection tracking regardless of whether the protocol supports connections. If a connection is allowed between a source and a target (for an ingress rule) or between a target and a destination (for an egress rule), all response traffic is allowed as long as the firewall's connection tracking state is active.

A firewall rule's tracking state is considered active if at least one packet is sent every 10 minutes.

When a fragmented connection is allowed through the firewall, Google Cloud uses connection tracking to allow only the first fragment of return traffic. To allow subsequent return fragments, you must add a firewall rule.

ICMP response traffic, such as "ICMP TYPE 3, DESTINATION UNREACHABLE", generated in response to an allowed TCP/UDP connection is allowed through the firewall. This behavior is consistent with RFC 792.

Multiple VPC

create several VPC networks and VM instances and test connectivity across networks. Specifically, you create two custom mode networks

- Create the managementnet network

- Create the privatenet network

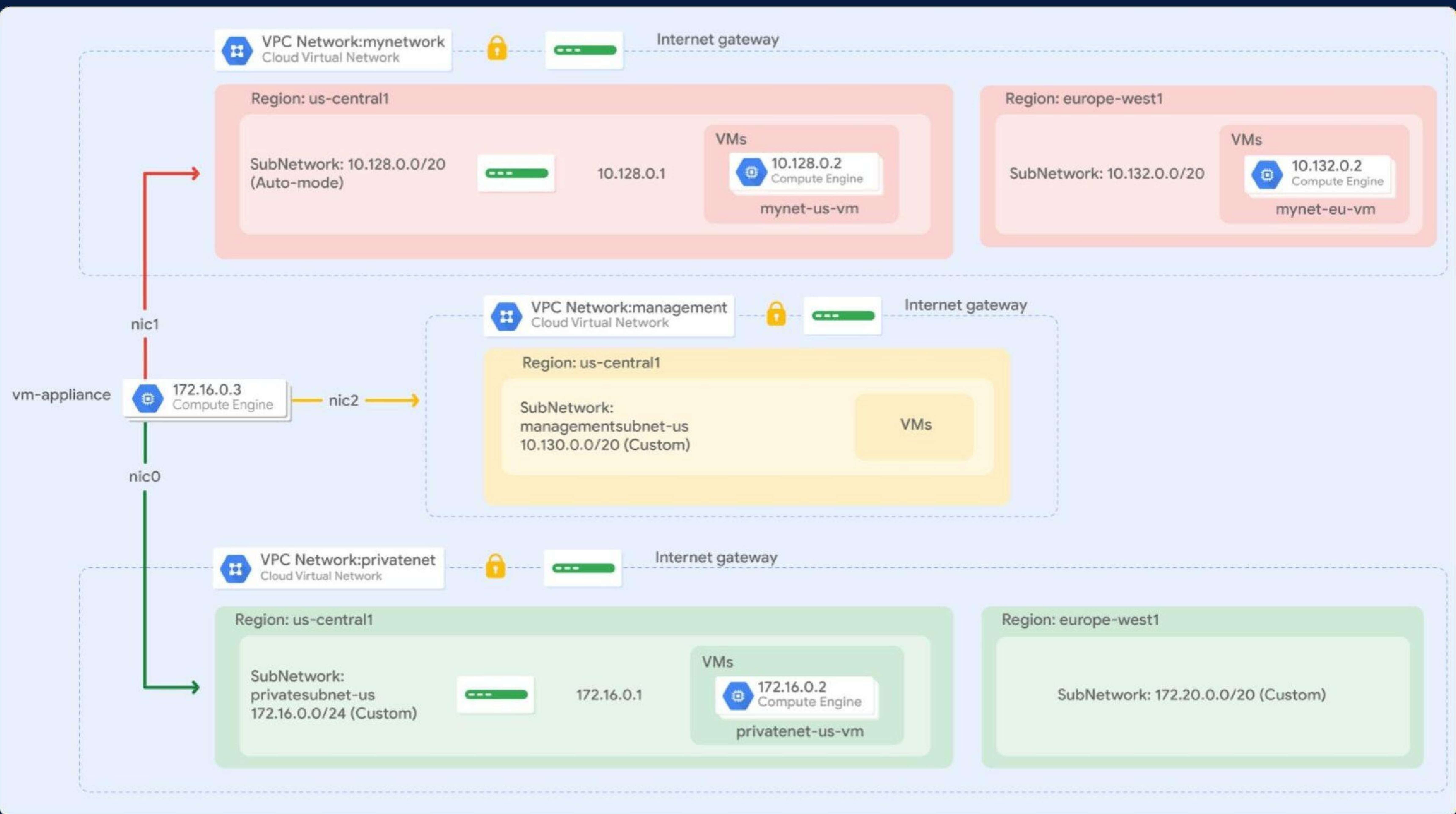
- Create the firewall rules for managementnet

- Create the firewall rules for privatenet

- Create the managementnet-us-vm instance

- Create the privatenet-us-vm instance

- Create a VM instance with multiple network interfaces



Hybrid Cloud

A hybrid cloud is a mixed computing environment where applications are run using a combination of computing, storage, and services in different environments—public clouds and private clouds, including on-premises data centers or “edge” locations. Hybrid cloud computing approaches are widespread because almost no one today relies entirely on a single public cloud.

hybrid cloud work

Hybrid clouds work by combining the resources and services from two or more separate computing environments. Hybrid cloud architectures require integration, orchestration, and coordination so you can share, shift, and synchronize information quickly.

Strong hybrid cloud networking is critical for a hybrid cloud deployment to function correctly. Interconnectivity between environments is typically established using a local area network (LAN), wide area network (WAN), virtual private network (VPN), and application programming interfaces (APIs).

Similar to other cloud computing architectures, hybrid cloud platforms leverage virtualization, containerization, and software-defined networking and storage technologies to abstract and aggregate resources. Dedicated management software allows organizations to allocate resources and enable on-demand provisioning to different environments.

How to set up a hybrid cloud

Workloads, infrastructure, and processes are unique. As a result, your hybrid cloud strategy will ultimately be defined around your business requirements. What works for one organization may not work for you.

When you build your hybrid cloud strategy, you will need to create a plan to:

Define your vision. Create a vision statement that evaluates your current approach and environments to help you define what you want to optimize and how a hybrid cloud deployment will enable you to reach your goals.

Assess workloads.

Decide the workloads that will run in or be migrated to each environment. Where workloads run can have a huge impact on the success of your hybrid cloud strategy. Putting a workload in the right place can make it more effective and efficient, while the wrong one can complicate your deployment.

Identify patterns.

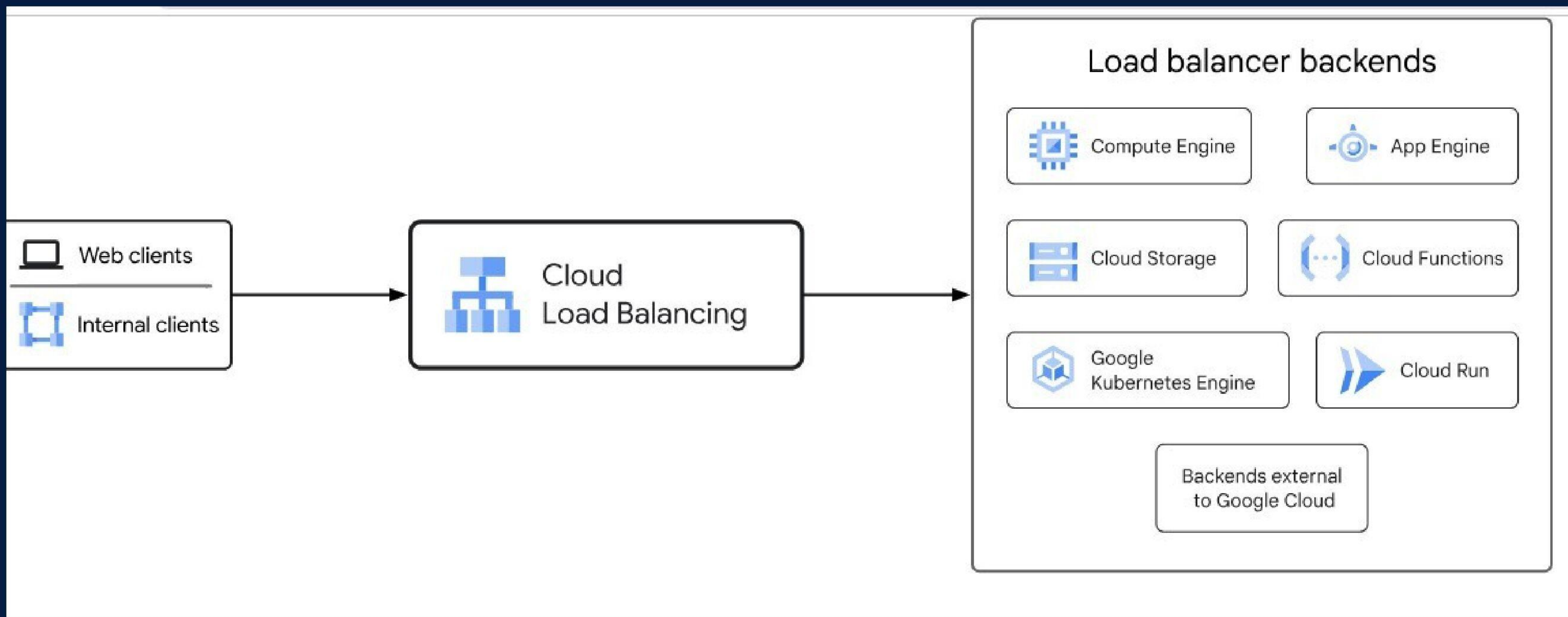
After you prioritize your workloads, you can evaluate which architecture patterns best match your requirements. Although your workloads are unique, there are common hybrid cloud architecture patterns you can use to help you make the most of your environments.

Choose technology and network topology.

Once you have decided on the architecture patterns for your workloads, decide which common network topologies and technologies will allow you to connect your environments in a secure and reliable manner.

Different Options of Load Balancing

A load balancer distributes user traffic across multiple instances of your applications. By spreading the load, load balancing reduces the risk that your applications experience performance issues. Google's Cloud Load Balancing is built on reliable, high-performing technologies such as Maglev, Andromeda, Google Front Ends, and Envoy—the same technologies that power Google's own products.

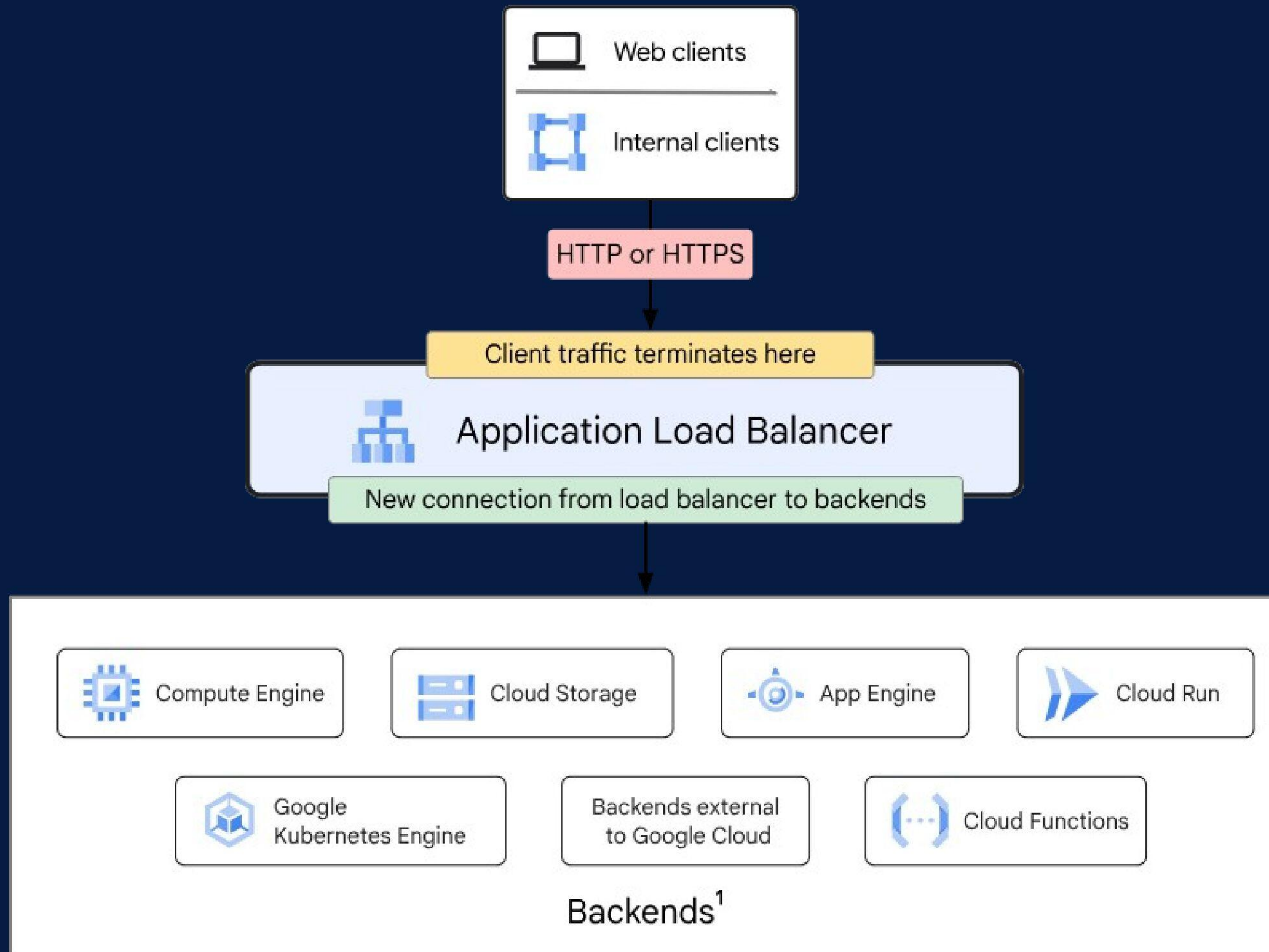


Types of load balancers

Cloud Load Balancing offers two types of load balancers: Application Load Balancers and Network Load Balancers.

Application Load Balancers

Application Load Balancers are proxy-based Layer 7 load balancers that enable you to run and scale your services behind a single IP address. The Application Load Balancer distributes HTTP and HTTPS traffic to backends hosted on a variety of Google Cloud platforms—such as Compute Engine, Google Kubernetes Engine (GKE), and Cloud Run—as well as external backends outside Google Cloud.



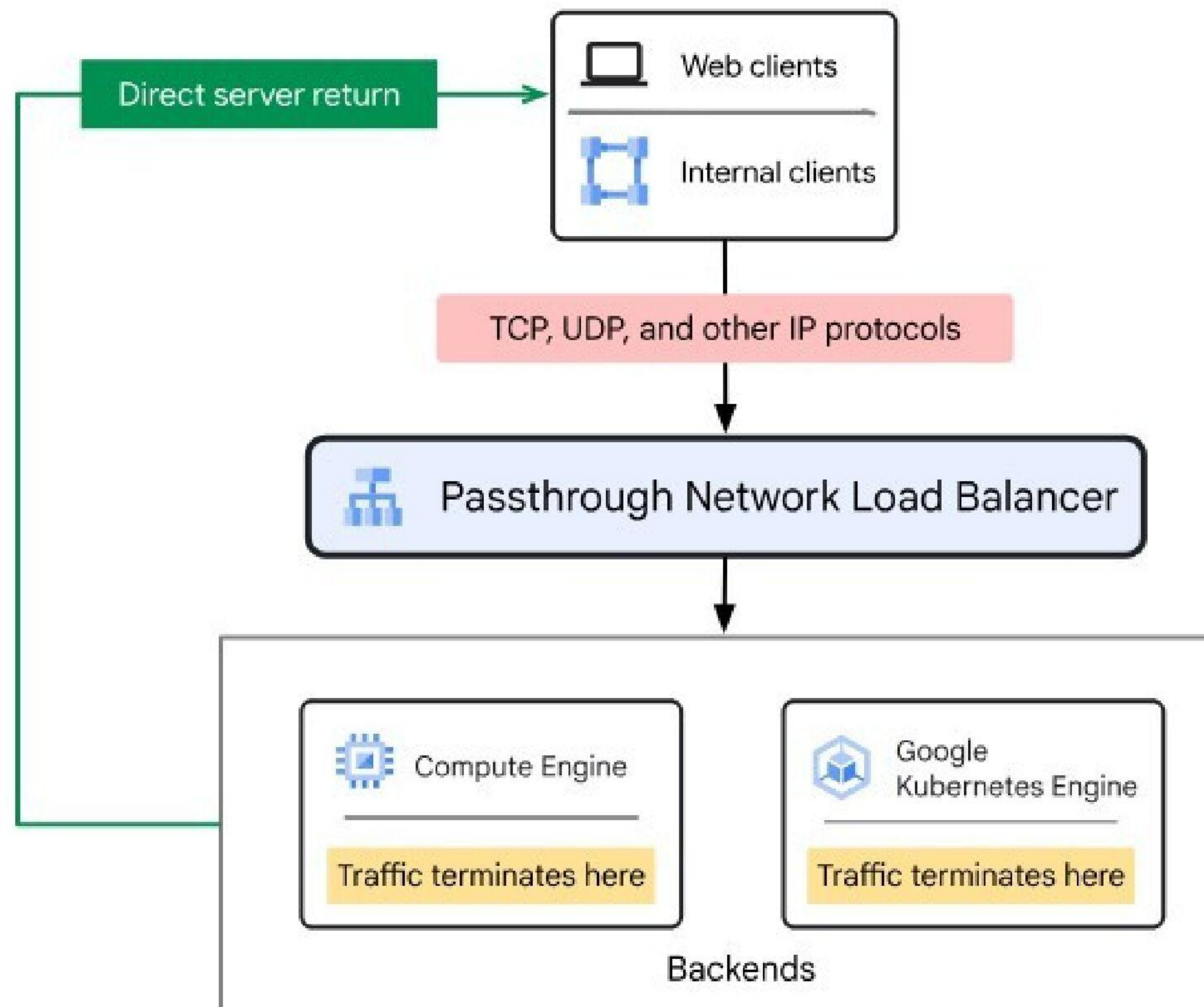
¹ Backend support differs depending on the deployment mode of the load balancer (internal or external, global or regional).

Network Load Balancers

Network Load Balancers are Layer 4 load balancers that can handle TCP, UDP, or other IP protocol traffic. These are available as either proxy load balancers or passthrough load balancers.

You can pick a load balancer depending on the needs of your application and the type of traffic that it needs to handle. Choose a proxy Network Load Balancer if you want to configure a reverse proxy load balancer with support for advanced traffic controls and backends on-premises and in other cloud environments.

Choose a passthrough Network Load Balancer if you want to preserve the source IP address of the client packets, you prefer direct server return for responses, or you want to handle a variety of IP protocols such as TCP, UDP, ESP, GRE, ICMP, and ICMPv6 .



Passthrough Network Load Balancer architecture.

Thank
You!